

Dimensional Reduction for Paraxial Beams with Thermally Mediated Nonlocal Nonlinearity

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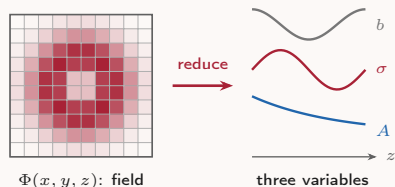
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1. **Select an ansatz.** A trial field with a few parameters, e.g. a width and a phase curvature.
 2. **Integrate the action** over the transverse plane, obtaining a Lagrangian in those parameters.
 3. **Apply the variational principle.** The Euler–Lagrange equations in the parameters are a few ordinary differential equations.
- **Advantage:** a model with few variables and an **effective potential**. **Limitation:** the transverse profile is prescribed by the ansatz.



Established applications: pulses in optical fibres,
[D. Anderson, *Phys. Rev. A* **27**, 3135–3145 (1983)]
[B. A. Malomed, *Progress in Optics* (2002)]
trapped condensates and the nonpolynomial Schrödinger equation,
[V. M. Pérez-García et al., *Phys. Rev. Lett.* **77**, 5320–5323 (1996)]
[L. Salasnich et al., *Phys. Rev. A* **65**, 043614 (2002)]
beams in graded-index media.
[FL et al., *Nanophotonics* **14**, 805–813 (2025)]

1) Solitons in multimode nonlinear fibres

$$i \partial_z \Phi = \left[-\frac{1}{2} \nabla_{\perp}^2 \underbrace{-\frac{1}{2} \partial_t^2}_{\text{anomalous dispersion}} + \underbrace{W(x, y)}_{\text{graded index}} + \underbrace{g |\Phi|^2}_{\text{Kerr}} \right] \Phi, \quad \Phi = \Phi(x, y, z, t).$$

- ▶ A short pulse in a **graded-index fibre**: a $(3+1)$ -D nonlinear Schrödinger equation.

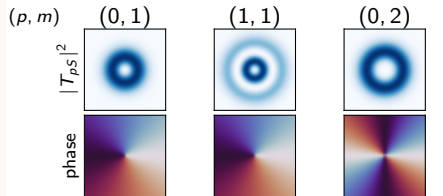
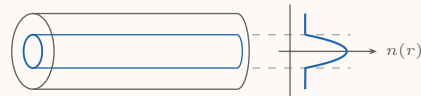
[FL et al., *Nanophotonics* 14, 805–813 (2025)]

- ▶ Φ is the envelope; lengths in the waist w_0 , z in $L_D = kw_0^2$, t in $T_0 = \sqrt{|\beta_2| L_D}$: dispersion and diffraction share L_D .

- ▶ For $g < 0$, formally the **Gross–Pitaevskii equation** of an attractive condensate, with z in the role of time: the **nonpolynomial Schrödinger approach** applies.

[L. Salasnich et al., *Phys. Rev. A* 65, 043614 (2002)]

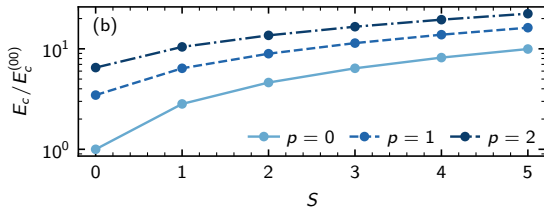
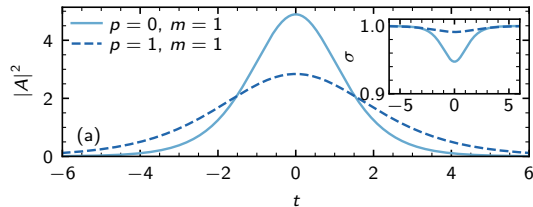
- ▶ **Ansatz**: one Laguerre–Gauss mode, $\Phi = A T_{ps}(r; \sigma) e^{im\theta} / \sqrt{2\pi}$, so $|A|^2$ is the power; A and σ depend on z and t .



The nonpolynomial Schrödinger equation and its solitons

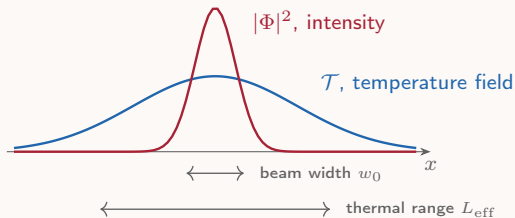
$$i \partial_z A = -\frac{1}{2} \partial_t^2 A + \xi_{pS} \frac{1 + \frac{3}{2} g_{pS} |A|^2}{\sqrt{1 + g_{pS} |A|^2}} A$$

- ▶ Substitute the width back: **one (1 + 1)-D equation** for the pulse.
- ▶ **(a)** Solitons in **closed form**; the width dips at the peak of the pulse (inset).
- ▶ **(b)** Solitons exist up to a **critical energy** E_c , larger for higher modes.
- ▶ **Vakhitov–Kolokolov**: for solitons $A = a(t) e^{-i\kappa z}$, stable where $dE/d\kappa < 0$.
- ▶ **Experiment**: multimode solitons observed in a graded-index fibre, femtosecond pulses at telecom wavelength over many dispersion lengths.
[W. H. Renninger et al., Nat. Commun. 4, 1719 (2013)]



2) Beams in a thermal medium: the index change spreads

- ▶ **Thermal effects.** Absorption heats the medium; the heat diffuses, following the **Fourier heat equation**.
- ▶ $\Delta n = \beta \mathcal{T}$, $\beta = dn/d\mathcal{T}$ the **thermorefractive index**, large in lead glass, silicon, liquid crystals: the index follows $\mathcal{T}$ over a **surrounding region**.
- ▶ A nonlocal response **arrests collapse**; a long enough range **stabilises ring beams**.
[O. Bang et al., Phys. Rev. E 66, 046619 (2002)]
[A. I. Yakimenko et al., Phys. Rev. E 71, 065603 (2005)]
- ▶ **Realised experimentally** with beams in lead glass and in cells of nematic liquid crystals: self-trapped beams and vortex solitons.
[C. Rotschild et al., Phys. Rev. Lett. 95, 213904 (2005)]
[C. Conti et al., Phys. Rev. Lett. 92, 113902 (2004)]



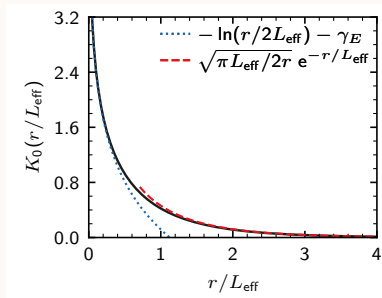
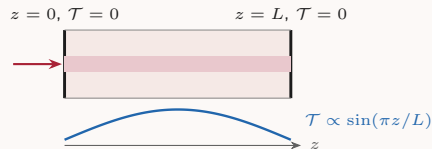
$$(-\nabla_{\perp}^2 + L_{\text{eff}}^{-2}) \mathcal{T} = \eta |\Phi|^2 \quad \Longrightarrow \quad \mathcal{T} = \eta G * |\Phi|^2,$$

$$G(r) = \frac{1}{2\pi} K_0\left(\frac{r}{L_{\text{eff}}}\right), \quad L_{\text{eff}} = \frac{L}{\pi}.$$

- **Steady heat equation**, absorbed power $\alpha_0 |\Phi|^2$ as source:
 $\eta = \alpha_0 / (\rho c_p \kappa)$ (density, specific heat, diffusivity).
- Facets at **fixed temperature**: the longitudinal mode $\sin(\pi z/L)$ adds the **screening** term L_{eff}^{-2} .
- The kernel K_0 is **logarithmic** below L_{eff} and decays **exponentially** above it; reorientational and electrostrictive nonlocalities obey the same screened Poisson equation, each with its own L_{eff} .

[A. Alberucci et al., Phys. Rev. A 91, 013841 (2015)]

[C. Conti et al., Phys. Rev. Lett. 95, 183902 (2005)]



The model: a paraxial beam in a graded-index medium

$$i \partial_z \Phi = \left(-i \frac{\alpha_0}{2} - \frac{1}{2} \nabla_{\perp}^2 + \frac{\Omega^2 r^2}{2} + \gamma \Theta[|\Phi|^2] \right) \Phi,$$

$$\Theta[n] = G * n.$$

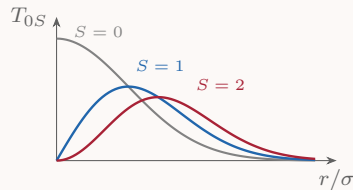
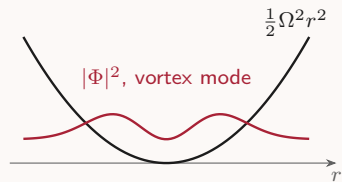
- Φ is the envelope; lengths in the waist w_0 , z in the diffraction length $L_D = kw_0^2$, k the wavenumber; power $P = \int |\Phi|^2 d^2 \mathbf{r}$.

- $\gamma \propto -\beta \eta$ ties the nonlinearity to the absorption: $\gamma > 0$ **defocusing** ($\beta < 0$), $\gamma < 0$ **focusing**.
[FL et al., Front. Phys. 21, 102203 (2026)]

- **Ansatz:** one annular mode with a radial chirp,

$$\Phi(r, \theta, z) = A(z) T_{pS}(r; \sigma(z)) \exp\left[i \frac{b(z)}{2} r^2\right] e^{im\theta}.$$

- $\xi_{pS} = S + 2p + 1$, with $S = |m|$ the charge and p the radial index.



Annular modes ($p = 0$): one ring; σ sets its radius.

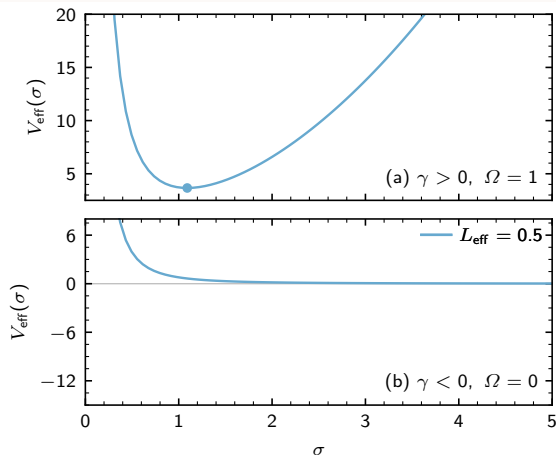
$$\xi_{pS} \ddot{\sigma} = -\frac{dV_{\text{eff}}}{d\sigma}, \quad V_{\text{eff}}(\sigma) = \frac{\xi_{pS}}{2} \left(\frac{1}{\sigma^2} + \Omega^2 \sigma^2 \right) + \frac{\gamma |A|^2}{4\pi} \mathcal{I}_{pS}(\sigma).$$

$$\mathcal{I}_{pS}(\sigma) = \int d^2\mathbf{r} \, \rho_{pS} \Theta[\rho_{pS}] = 2\pi \int_0^\infty dq \, q \frac{L_{\text{eff}}^2}{1 + L_{\text{eff}}^2 q^2} \tilde{\rho}_{pS}(q; \sigma)^2.$$

- ▶ Three Euler–Lagrange equations, for A , σ , b . The amplitude decouples, $|A|^2 = |A_0|^2 e^{-\alpha_0 z}$, and the chirp is $b = \dot{\sigma}/\sigma$.
- ▶ What remains is **one** equation: a **particle of mass ξ_{pS}** in the well V_{eff} , which depends on z only through $|A|^2$.
- ▶ **Loss** enters **phenomenologically**, through the decay of $|A|^2$ in V_{eff} alone.
- ▶ $\mathcal{I}_{pS}$: the **nonlocal interaction energy** of the profile $\rho_{pS} = T_{pS}^2$; $\tilde{\rho}_{pS}$ its Hankel transform.
- ▶ One radial integral. For the annular modes ($p = 0$) it is in **closed form**: for $S = 0$, $\mathcal{I}_{00} = -\pi e^u \text{Ei}(-u)$ with $u = \sigma^2/2L_{\text{eff}}^2$.
- ▶ Local limit $\sigma \gg L_{\text{eff}}$: $\mathcal{I}_{0S} \propto L_{\text{eff}}^2/\sigma^2$, the **Kerr** scaling.

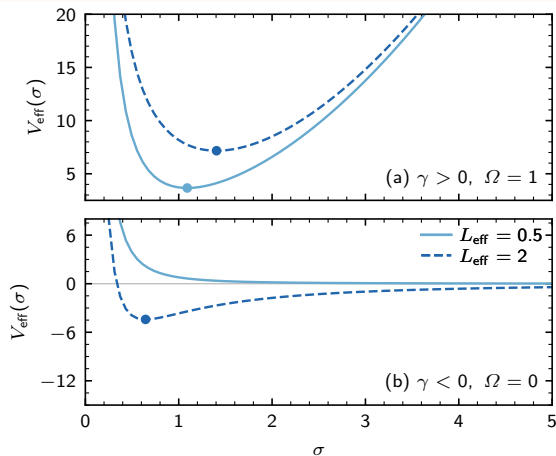
Stationary states: L_{eff} sets the width, and there is no collapse

- ▶ V_{eff} for an annular mode with $S = 2$, at three nonlocal lengths; dots mark the minima.
- ▶ (a) **Defocusing in the trap.** The stationary width grows with L_{eff} : the beam **broadens**.
- ▶ (b) **Focusing without trap.** Diffraction exceeds the logarithm as $\sigma \rightarrow 0$: **no collapse**.



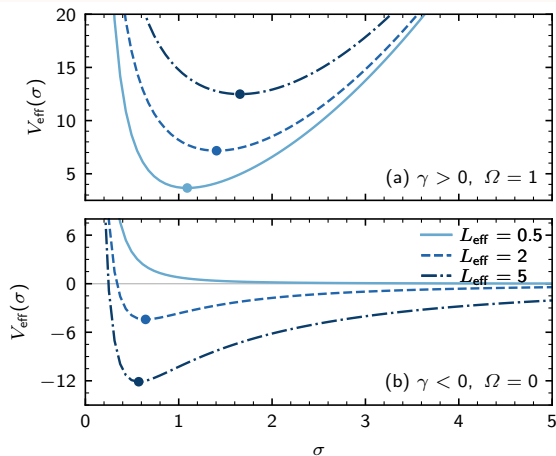
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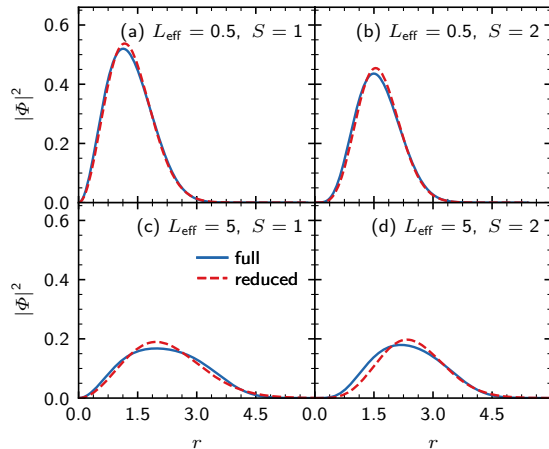


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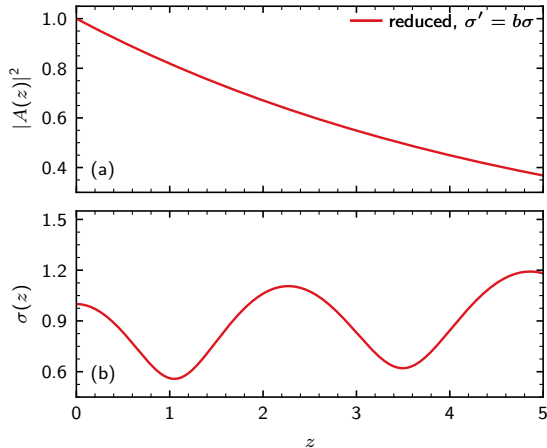
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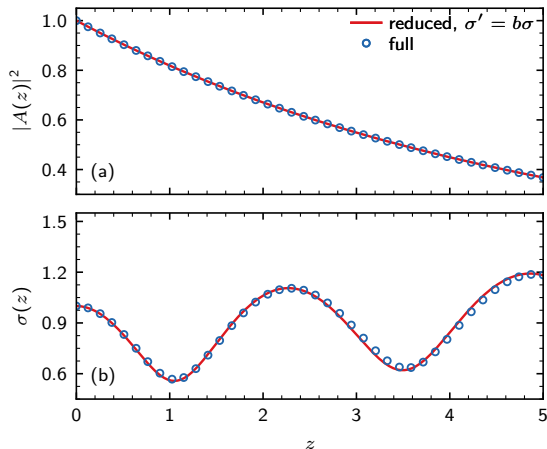
- **Full model:** imaginary time at fixed charge m .
Reduced: the mode at the minimum of V_{eff} .



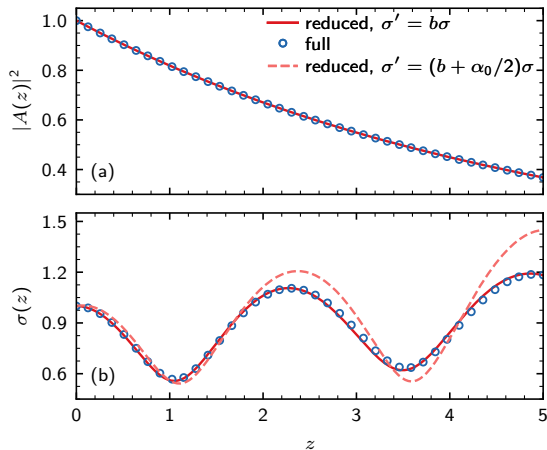
- ▶ An $m = 1$ vortex launched at the linear-mode width: focusing, $L_{\text{eff}} \gg \sigma$, lossy, trapped.
- ▶ **(b)** Loss in the amplitude alone: the reduced width follows the full run within 4%.
- ▶ Reference: a 1 W vortex ring in lead glass, $L_{\text{eff}}/w_0 \approx 40$ as here, $\alpha_0 \approx 0.015$, no trap.
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Conclusions

1. Thermal nonlocality between fixed-temperature facets, at weak absorption: a K_0 **kernel** of range $L_{\text{eff}} = L/\pi$.
2. One annular mode with a chirp: a **reduced equation for the width**; the nonlocality enters through $\mathcal{I}_{pS}(\sigma)$ alone.
3. In the simulated regimes the reduced width **follows the full model closely**; azimuthal stability is not captured.

Based on

[FL et al., Nanophotonics 14, 805–813 (2025)]

[FL et al., Front. Phys. 21, 102203 (2026)]



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Thank you for your attention

Questions?

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Part A

Additional material

The radial profile is $T_{pS}(r; \sigma) = \frac{C_{pS}}{\sigma} \left(\frac{r}{\sigma}\right)^S L_p^S\left(\frac{r^2}{\sigma^2}\right) e^{-r^2/2\sigma^2}$, with $C_{pS} = \sqrt{2p!/(p+S)!}$ so that $\int_0^\infty r dr T_{pS}^2 = 1$. Inserting the ansatz in the lossless action, integrating over the transverse plane and dividing by 2π gives

$$\mathcal{L} = iA^* \dot{A} - \frac{\xi_{pS}}{2} [\sigma^{-2} + (\Omega^2 + b^2 + \dot{b}) \sigma^2] |A|^2 - \frac{\gamma}{4\pi} |A|^4 \mathcal{I}_{pS}(\sigma),$$

The three equations

$$i\dot{A} = \frac{\xi_{pS}}{2} [\sigma^{-2} + (\Omega^2 + b^2 + \dot{b}) \sigma^2] A + \frac{\gamma}{2\pi} \mathcal{I}_{pS} |A|^2 A,$$

$$0 = \xi_{pS} [\sigma^{-3} - (\Omega^2 + b^2 + \dot{b}) \sigma] - \frac{\gamma}{4\pi} |A|^2 \mathcal{I}'_{pS},$$

$$\frac{d}{dz} (\sigma^2 |A|^2) = 2b \sigma^2 |A|^2.$$

The second is algebraic in σ , so the width condition holds at every z ; the third comes from $\partial \mathcal{L} / \partial \dot{b} = -\frac{\xi_{pS}}{2} \sigma^2 |A|^2$.

Loss. $\Phi = e^{-\alpha_0 z/2} \Psi$ turns the lossy equation into the conservative one for Ψ with $\gamma \rightarrow \gamma e^{-\alpha_0 z}$, which the reduction admits unchanged: $|A_\Psi|^2$ is conserved, so $\dot{\sigma} = b\sigma$ still holds and V_{eff} inherits the decaying product $\gamma |A|^2$.

Reduction to one equation

With $|A|^2$ constant the third equation is $\dot{\sigma} = b\sigma$, hence $b^2 + \dot{b} = \ddot{\sigma}/\sigma$. Substituting in the second,

$$\xi_{pS} \ddot{\sigma} = \xi_{pS} (\sigma^{-3} - \Omega^2 \sigma) - \frac{\gamma |A|^2}{4\pi} \mathcal{I}'_{pS} = -\frac{dV_{\text{eff}}}{d\sigma}.$$

The amplitude equation then only sets the phase of A . For $\gamma = 0$ this is the Ermakov equation $\ddot{\sigma} = \sigma^{-3} - \Omega^2 \sigma$ of Gaussian-beam optics.

[H. R. Lewis, *Phys. Rev. Lett.* **18**, 510–512 (1967)]

Closed forms for $\mathcal{I}_{0S}$ and its two limits

Annular modes: $\rho_{0S}(r) = \frac{2r^{2S}}{\sigma^{2S+2}S!} e^{-r^2/\sigma^2}$, whose

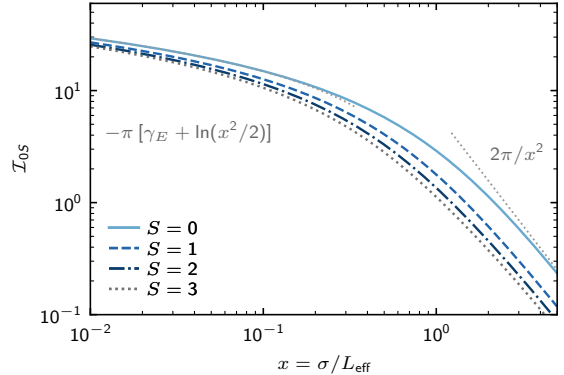
Hankel transform is the confluent hypergeometric function

$$\tilde{\rho}_{0S}(q; \sigma) = {}_1F_1\left(1 + S; 1; -\frac{q^2\sigma^2}{4}\right).$$

Gaussian mode ($S = 0$), with $u = \sigma^2/2L_{\text{eff}}^2$:

$$\mathcal{I}_{00} = -\pi e^u \text{Ei}(-u), \quad \frac{d\mathcal{I}_{00}}{d\sigma} = \frac{\sigma}{L_{\text{eff}}^2} \mathcal{I}_{00} - \frac{2\pi}{\sigma}.$$

- $\sigma \ll L_{\text{eff}}$: $\mathcal{I}_{00} \simeq -\pi[\gamma_E + \ln(x^2/2)]$, $x = \sigma/L_{\text{eff}}$ as on the panel: a logarithm, weaker than diffraction (σ^{-2}), so $V_{\text{eff}} \rightarrow +\infty$ as $\sigma \rightarrow 0$ for either sign of γ .
- $\sigma \gg L_{\text{eff}}$: $\mathcal{I}_{0S} \rightarrow 2\pi L_{\text{eff}}^2 c_S/\sigma^2$, $c_S = \frac{(2S)!}{4^S(S!)^2}$, from Parseval's identity: the local Kerr limit, with the σ^{-2} law of the local NPSE.
- S orders the curves: the $S = 0$ curve lies above the others over the range shown, and no single power law holds for all σ .



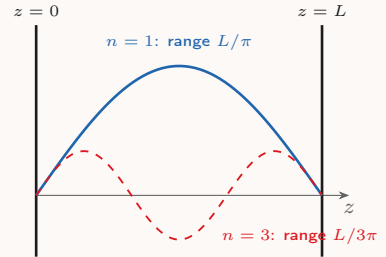
Why $L_{\text{eff}} = L/\pi$

- The facet conditions $\mathcal{T}(z = 0) = \mathcal{T}(z = L) = 0$ are met by the expansion

$$\mathcal{T}(\mathbf{r}, z) = \sum_{n \geq 1} \mathcal{T}_n(\mathbf{r}) \sin \frac{n\pi z}{L}, \quad \left(-\nabla_{\perp}^2 + \frac{n^2\pi^2}{L^2}\right) \mathcal{T}_n = \eta s_n(\mathbf{r}),$$

with $s_n = \frac{2}{L} \int_0^L |\Phi|^2 \sin \frac{n\pi z}{L} dz$ the projection of the source.

- Each $\mathcal{T}_n$ is a screened Poisson response of range $L/(n\pi)$. For a source independent of z (weak absorption, $\alpha_0 L \ll 1$, and a stationary beam) $s_n = 4|\Phi|^2/n\pi$ for odd n and 0 for even n .
- Only the $n = 1$ term is kept, the one of longest range, its weight $4/\pi$ absorbed into γ : $L_{\text{eff}} = L/\pi$. The terms $n \geq 3$ carry weights $1/n$ of it and ranges $L/n\pi$; they modify the kernel at $r \lesssim L/3\pi$ and are neglected here.
- Strong absorption changes the longitudinal profile and with it the value of L_{eff} ; the full three-dimensional Green's function is available.
[N. Ghofraniha et al., Phys. Rev. Lett. 99, 043903 (2007)]
[A. Alberucci et al., Phys. Rev. A 82, 023806 (2010)]
[G. Marcucci et al., Adv. Phys. X 4, 1662733 (2019)]

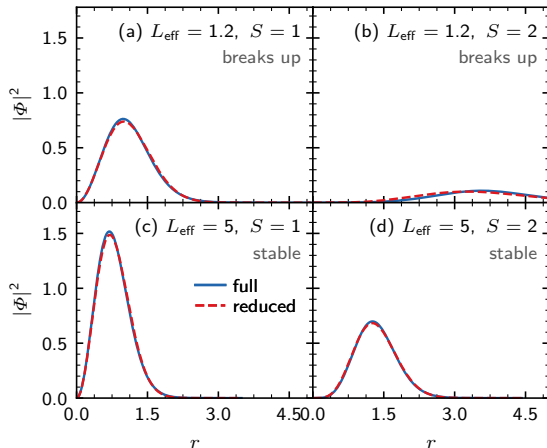


Longitudinal modes between fixed-temperature facets; the higher modes are screened over shorter transverse distances.

- **Full model:** the symmetric stationary state of the radial equation at $P = 2\pi$ (imaginary time, then Newton; residual 10^{-13}). **Reduced:** the annular mode at the minimum of V_{eff} .
- The widths agree in all four panels: 0.2%, 2.5%, 0.1%, 0.06%. **The ansatz is as accurate as in the trap.**
- The reduction does not describe the **azimuthal stability** of these states: at $L_{\text{eff}} = 1.2$ both charges break up, at $L_{\text{eff}} = 5$ both are stable.
- Two-dimensional imaginary time started from an azimuthally unstable state drifts away from it; the focusing profiles of Fig. 3(b–d) in the Frontiers of Physics manuscript are of that kind, not stationary states.

[FL et al., *Front. Phys.* **21**, 102203 (2026)]

[W. Bao et al., *SIAM J. Sci. Comput.* **25**, 1674–1697 (2004)]



- **Full model:** split-step Fourier propagation; diffraction and trap in the linear step, Θ by convolution with the K_0 kernel at every axial step; stationary states at fixed charge m by imaginary time.

[W. Krolikowski et al., Phys. Rev. E 64, 016612 (2001)]

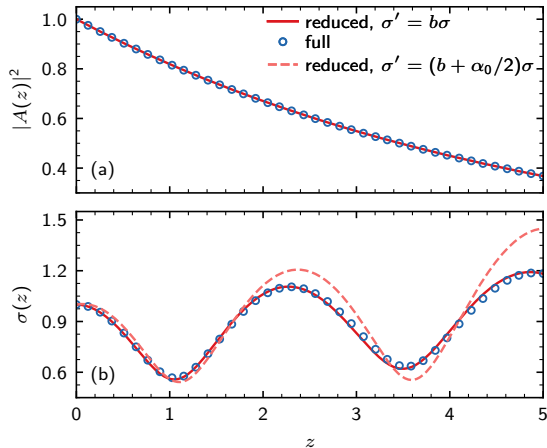
[A. I. Yakimenko et al., Phys. Rev. E 71, 065603 (2005)]

- **Reduced model:** Runge–Kutta integration of $\xi_{pS}\ddot{\sigma} = -dV_{\text{eff}}/d\sigma$ with $|A|^2 = |A(0)|^2 e^{-\alpha_0 z}$; $d\mathcal{I}_{0S}/d\sigma$ from the Hankel integral.
- Keeping $d(\sigma^2|A|^2)/dz = 2b\sigma^2|A|^2$ with a decaying $|A|^2$ gives $\dot{\sigma} = (b + \alpha_0/2)\sigma$, that is

$$\xi_{pS}\ddot{\sigma} = -\frac{dV_{\text{eff}}}{d\sigma} + \xi_{pS}\left(\alpha_0\dot{\sigma} - \frac{\alpha_0^2}{4}\sigma\right) :$$

a negative friction: it amplifies the breathing and lifts the width 22% above the full model at $z = 5$. With the loss in the amplitude alone the two models agree within 4% throughout.

- L_{eff} is a free parameter here: read as L/π the sample would be $126 w_0 = 0.2 L_D$ for the reference beam, shorter than the window; in the experiment the lateral wall sets L_{eff} .



Linear stability of the full-model state $\phi(r)e^{im\theta}$ under $u(r)e^{i\ell\theta} + v^*(r)e^{-i\ell\theta}$: one eigenvalue problem per ℓ . $\text{Re } \lambda > 0$: a modulation with ℓ lobes grows as $e^{\lambda z}$. Stable: $\text{Re } \lambda < 10^{-5}$ for $\ell \leq 6$, the untrapped $\ell = 1$ translation mode excluded (its rate scales with the grid step). Two-dimensional split-step check, trapped $m = 2$ at $L_{\text{eff}} = 0.5$, seeded $\ell = 2$: predicted rate 0.163, fitted 0.161.

State	m	Ω	L_{eff}	Verdict (rate per diffraction length)
stationary profiles (a), (c)	1	1	0.5, 5	stable
stationary profiles (b), (d)	2	1	0.5, 5	splits into two $m = 1$ vortices, $\ell = 2$: 0.163, 0.037
propagation run, launch and end power	1	1	40	stable
A4 (a), (b)	1, 2	0	1.2	breaks up: $\ell = 2$ at 0.69 ($m = 1$), $\ell = 3$ at 0.33 ($m = 2$)
A4 (c), (d)	1, 2	0	5	stable
V_{eff} wells, panel (b)	2	0	2, 5	breaks up ($\ell = 2$: 0.31); stable

Every $m = 1$ state of the main frames is stable. The $m = 2$ states need **strong nonlocality**: in the trap they split, four times slower at $L_{\text{eff}} = 5$ than at 0.5; without the trap, at these powers, the threshold lies between $L_{\text{eff}} = 2$ and 5 for $m = 2$ and between 1.2 and 5 for $m = 1$, as in the nonlocal stabilisation of vortex solitons.

[A. I. Yakimenko et al., *Phys. Rev. E* **71**, 065603 (2005)]

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